

Uniqueness of the Differential Structures on $\mathbb{R}$ and $\mathbb{R}^2$

Zheng(Peter) Jiang

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Introduction: The Big Question and Established Results

Question

What can we say about the equivalence classes of smooth atlases (or smooth structures) on $\mathbb{R}^n$ when equipped with the standard topology?

(For a topological manifold X , and two of its smooth atlases A, B , we say $A \sim B \iff$ there exists a diffeomorphism $f : (X, A) \rightarrow (X, B)$)

Answer

For $n \neq 4$, we only have the standard structure. But $\mathbb{R}^4$ has uncountably many exotic structures.

Remark

Simon Donaldson won a Fields Medal in 1986 for his work in gauge theory and smooth 4-manifolds. His work showed, among other things, that there are uncountably many structures on $\mathbb{R}^4$

Today's focus

Uniqueness results for $\mathbb{R}, \mathbb{R}^2$

- Existence of a chain-like atlas
- Gluing the chain

Structure on $\mathbb{R}$: Existence of the chain-like atlas

Goal

Given an atlas A of $\mathbb{R}^2$, we wish to find a diffeomorphism from $(\mathbb{R}^2, A)$ to $(\mathbb{R}^2, \mathcal{ST})$

Definition (Chain-like Atlas)

An atlas B is *chain-like* if $B = \{(a_i, b_i), \phi_i\}_{i \in S}$, where $S = \mathbb{Z}, \mathbb{N}, -\mathbb{N}$, or $\{-n, -n+1, \dots, m-1, m\}$, and $\forall i, a_i < b_{i-1} < a_{i+1} < b_i$. In other words we have $(a_i, b_i) \cap (a_j, b_j) \cap (a_k, b_k) = \emptyset$ whenever $i \neq j \neq k \neq i$.

Lemma (Zorn's lemma)

Let $(P, \leq)$ be a partially ordered set such that every chain $C \subseteq P$ has an lower bound in P . Then P contains at least one minimal element.

Desired partial order

Define partial order $\leq$ on $S = \{X \in 2^A \mid \bigcup_{(U, \phi) \in X} U = \mathbb{R}\}$. For $P, Q \in S$ we say $Q \leq P \iff Q \subset P$. Minimal element (atlas) denoted as $\hat{A}$, we then have $(\mathbb{R}, A) \cong (\mathbb{R}, \hat{A})$.

Definition (Joinable charts)

For a 1-manifold M , and charts (U_1, f_1) , (U_2, f_2) , we say two charts are *joinable* if $f_1(U_1) = I_1$, $f_2(U_2) = I_2$, $f_1(U_1 \cap U_2) = f_2(U_1 \cap U_2) = I_1 \cap I_2$ for some open intervals I_1, I_2 with $I_1 \cap I_2 \neq \emptyset$.

Lemma (Existence of the joined chart)

For two joinable charts (U_1, f_1) , (U_2, f_2) , there exists a joined chart $(U_1 \cup U_2, f)$ which satisfies the following three properties

- (1). $f(U_1) = f_1(U_1)$, $f(U_2) = f_2(U_2)$
- (2). $(U_1 \cup U_2, f)$ is compatible with the two charts.
- (3). $f|_{U_1 \setminus U_2} = f_1|_{U_1 \setminus U_2}$, $f|_{U_2 \setminus U_1} = f_2|_{U_2 \setminus U_1}$

Remark

Think of the joined chart as a smooth interpolation of two joinable charts, or view it as the gluing of two joinable charts

Structure on $\mathbb{R}$: Gluing the Chain

Lemma

Adjacent charts in the chain-like atlas can be transformed into joinable charts $\hat{A}$ by applying suitable trivial diffeomorphisms to their images $(\ln, \tan, \frac{1}{x}, \dots)$. Hence, eventually we can get a chain-like atlas B with adjacent charts all being joinable.

Lemma

There exist some atlas C with a single chart $(\mathbb{R}, F)$ with $F : \mathbb{R} \rightarrow \mathbb{R}$ a homeomorphism, compatible with the transformed atlas B . So we have $(\mathbb{R}, C) \cong (\mathbb{R}, B)$.

Proof.

Replace two joinable charts with their joined chart in the atlas, repeat the process, pass it to infinity. □

Conclusion

we have $(\mathbb{R}, C) \cong (\mathbb{R}, ST)$ by F^{-1} , hence $(\mathbb{R}, A) \cong (\mathbb{R}, \hat{A}) \cong (\mathbb{R}, B) \cong (\mathbb{R}, C) \cong (\mathbb{R}, ST)$.

- Smoothing theorems
- Smoothing Near Vertices
- Smoothing Near Edges
- Smoothing of Faces

Theorem (Handle Smoothing Theorem)

For a homeomorphism between two smooth triangulable surfaces.

- 1. We can isotope the homeomorphism into a diffeomorphism near each vertex.*
- 2. The isotopy can be extended to a neighbourhood of each edge.*
- 3. It can be extended further to each face.*

Theorem (THE THEOREM)

A homeomorphism between two smooth triangulable surfaces can be isotoped into a diffeomorphism.

Remark

$\mathbb{R}^2$ is triangulable and $\text{id} : (\mathbb{R}^2, A) \rightarrow (\mathbb{R}^2, B)$ for any pair of atlases A, B is a homeomorphism.

Remark

The theorem shows that smooth structures of the same manifold are unique not just up to diffeomorphism, but up to isotopy.

Theorem (Vertex Smoothing Theorem)

For a smooth surface (S, σ) , and a topological embedding $h_0 : (\mathbb{R}^2, ST) \rightarrow (S, \sigma)$, we can then isotope h_0 to a smooth embedding h_1 near the origin with $h_1|_{\mathbb{R}^2 \setminus U} = h_0|_{\mathbb{R}^2 \setminus U}$ for some larger neighbourhood U of the origin.

$$\begin{array}{ccccccccc}
 (\mathbb{R}^2) & \xrightarrow{i_3} & (B(1), \chi_5) & \xrightarrow{\phi^{-1}} & (\mathbb{R}^2, \chi_4) & \xrightarrow{q} & (\mathbf{T}, \chi_3) & \xrightarrow{i_2} & (\mathbf{T}_0, \chi_2) & \xrightarrow{i_1} & (\mathbb{R}^2, \chi_1) \\
 \downarrow G & & \downarrow \hat{G} & & \downarrow \hat{g} & & \downarrow g & & & & \downarrow h_0 \\
 (\mathbb{R}^2, ST) & \xrightarrow{i_3} & (B(1), ST) & \xleftarrow{\phi} & (\mathbb{R}^2, ST) & \xrightarrow{q} & (\mathbf{T}, ST) & & & & (S, \sigma)
 \end{array}$$

Structure on $\mathbb{R}^2$: Smoothing Near Vertices & Kirby Torus Trick

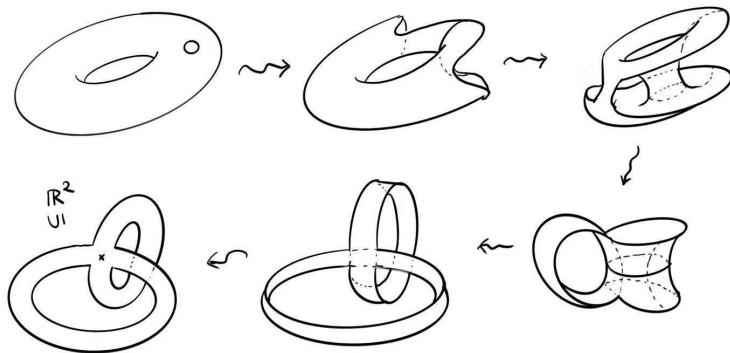


Figure: Immersion of the punctured torus into $\mathbb{R}^2$

Structure on $\mathbb{R}^2$: Smoothing Near Vertices & Kirby Torus Trick

Lemma

Given any atlas A on $\mathbf{T}_0$, there exists some open $U \subset \mathbf{T}_0$ such that $U \cong S^1 \times \mathbb{R}$ with $U \cup \{\star\}$ being a neighbourhood of $\{\star\}$ in $\mathbf{T}$.

Lemma

Every atlas on $\mathbf{T}_0$ can be extended to an atlas on $\mathbf{T}$

Theorem

Smooth structure on $\mathbf{T}$ is unique

$$\begin{array}{ccc} (\mathbf{T}, \chi_3) & \xrightarrow{i_2} & (\mathbf{T}_0, \chi_2) \xrightarrow{i_1} (\mathbb{R}^2, \chi_1) \\ \downarrow g & & \downarrow h_0 \\ (\mathbf{T}, \text{ST}) & & (S, \sigma) \end{array}$$

Structure on $\mathbb{R}^2$: Smoothing Near Vertices & Kirby Torus Trick

Theorem (Lift Existence Theorem)

Let $\hat{X}$ be a covering space of X , and $f : Y \rightarrow X$. Then f lifts to $\hat{f} : Y \rightarrow \hat{X}$ if and only if $f_*(\pi_1(Y, y_0)) \subset p_*(\pi_1(\hat{X}, \hat{x}_0))$

$$\begin{array}{ccc} (\mathbb{R}^2, \chi_4) & \xrightarrow{q} & (\mathbf{T}, \chi_3) \\ \hat{g} \downarrow & \searrow g \circ q & \downarrow g \\ (\mathbb{R}^2, \text{ST}) & \xrightarrow{q} & (\mathbf{T}, \text{ST}) \end{array}$$

Remark

There exists some $C > 0$, such that for all $(x, y) \in \mathbb{R}^2$, we have $|\hat{g}(x, y) - (x, y)| < C$.

Radial Reparameterization

$\phi : \mathbb{R}^2 \rightarrow B(1)$ by $\tanh$ on the radial component (for large r), but with $\phi = \text{id}$ on $B(\varepsilon)$.

$$\begin{array}{ccc} (B(1), \chi_5) & \xrightarrow{\phi^{-1}} & (\mathbb{R}^2, \chi_4) \\ \hat{G} \downarrow & & \downarrow \hat{g} \\ (B(1), \text{ST}) & \xleftarrow{\phi} & (\mathbb{R}^2, \text{ST}) \end{array}$$

Structure on $\mathbb{R}^2$: Smoothing Near Vertices & Kirby Torus Trick

$$\begin{array}{ccccccccc}
 (\mathbb{R}^2) & \xrightarrow{i_3} & (B(1), \chi_5) & \xrightarrow{\phi^{-1}} & (\mathbb{R}^2, \chi_4) & \xrightarrow{q} & (\mathbf{T}, \chi_3) & \xrightarrow{i_2} & (\mathbf{T}_0, \chi_2) & \xrightarrow{i_1} & (\mathbb{R}^2, \chi_1) \\
 \downarrow G & & \downarrow \hat{G} & & \downarrow \hat{g} & & \downarrow g & & & & \downarrow h_0 \\
 (\mathbb{R}^2, \text{ST}) & \xrightarrow{i_3} & (B(1), \text{ST}) & \xleftarrow{\phi} & (\mathbb{R}^2, \text{ST}) & \xrightarrow{q} & (\mathbf{T}, \text{ST}) & & & & (S, \sigma)
 \end{array}$$

Extension of $\hat{G}$

Extend $\hat{G}$ to G , where G is the identity outside of $B(1)$ (because of the reparameterization and the constant bound C). Then $h_0 \circ G^{-1}$ is the desired diffeomorphism.

Alexander's Disk Trick

$G_t|_{tB(1)}(x) = tG(x/t) \subset B(1)$, $G_t|_{\mathbb{R}^2 \setminus tB(1)} = \text{id}$ for $t \in (0, 1]$, $G_0 = \text{id}$. We have $G_1 = G$, hence $h_t = h_0 \circ G_t^{-1}$ gives the desired isotopy.

Structure on $\mathbb{R}^2$: Smoothing Near edges

Lemma

For an atlas A standard near $\partial D^1 \times \mathbb{R}$. We have $(D^1 \times \mathbb{R}, A) \cong (D^1 \times \mathbb{R}, ST)$, and the diffeomorphism can be chosen to be the identity near $\partial D^1 \times \mathbb{R}$.

Theorem (Edge Smoothing Theorem)

For a smooth surface (S, σ) and a topological embedding $h_0 : (D^1 \times \mathbb{R}, ST) \rightarrow (S, \sigma)$, with h_0 smooth near $\partial D^1 \times \mathbb{R}$. We can then isotope h_0 to a smooth embedding h_1 near $D^1 \times \{0\}$ with the isotopy being the identity near $\partial D^1 \times \mathbb{R}$, and $h_1|_{D^1 \times \mathbb{R} \setminus U} = h_0|_{D^1 \times \mathbb{R} \setminus U}$ for some larger neighbourhood U of $D^1 \times \{0\}$ (here we assume $D^1 = [-1, 1]$).

Notation

$$C = (D^1 \times D^1) \setminus (\{0\} \times \partial D^1)$$

$$\begin{array}{ccccccc} D^1 \times \mathbb{R} & \xrightarrow{i} & (C, \chi_2) & \xrightarrow{\phi^{-1}} & (D^1 \times \mathbb{R}, \chi_1) & \xrightarrow{h_0} & (S, \sigma) \\ \downarrow G & & \downarrow \hat{G} & & \downarrow g & & \\ (D^1 \times \mathbb{R}, ST) & \xrightarrow{i} & (C, ST) & \xleftarrow{\phi} & (D^1 \times \mathbb{R}, ST) & & \end{array}$$

Lemma

For an atlas A standard near $\partial D^1 \times \mathbb{R}$. We have $(D^2, A) \cong (D^2, ST)$, and the diffeomorphism can be chosen to the identity near S^1 .

Theorem (Face Smoothing Theorem)

For a smooth surface (S, σ) and a topological embedding $h_0 : (D^2, ST) \rightarrow (S, \sigma)$ with h_0 smooth near S^1 . We can then isotope h_0 to a smooth embedding h_1 on D^2 .